# Ramakrishna Pinninti

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## Personal Profile

I am an AI/NLP Research Engineer at **GREYC (CNRS / Université de Caen Normandie)**, working on the ANR-funded national project **2E2MR**, with an **Honours Level 9 Master's by Research** from Munster Technological University, specialising in Large Language Models, Natural Language Processing, and Information Retrieval. I hold **competitive EU/MSCA and Science Foundation Ireland research funding** and bring industry experience in cloud-scale data engineering at Accenture. I serve as a university lecturer at MTU and contribute to international AI research collaborations spanning Ireland, France, India, and the USA, and I am technically versatile across Python, SQL, AWS, GCP, and LLM frameworks,with a strong record of translating complex computational ideas into practical, deployable solutions.

## Education

**Master's by Research in Computer Science** — AI, NLP & Information Retrieval *2023 – 2026*
Munster Technological University (MTU), Ireland | Supervisor: Dr. Mohammed Hasanuzzaman, ADAPT Centre

**Bachelor of Computer Science** *2019 – 2023*
Indian Institute of Technology, Patna, India

## Work Experience

**Research Engineer** *Oct 2026 – present*
*GREYC Laboratory (UMR 6072) — CNRS & Université de Caen Normandie (UNICAEN), Caen, France*

- Research engineer on the **ANR 2E2MR project** (ANR-24-CE38-3598), a **French national interdisciplinary project** (computer science × cognitive/sport science) with Univ. Rouen Normandie, Savoie Mont Blanc, Lille and ETH Zurich.
- Building an **end-to-end LLM pipeline** for elite athletes' self-confrontation interviews: **Whisper** speech-to-text with timestamp alignment to video, followed by structured extraction of **hexadic signs** (course-of-experience framework) using locally hosted open-weight LLMs (**Ollama**; Qwen2.5, Llama 3) for data privacy.
- Applying **Retrieval-Augmented Generation (RAG)**: indexing framework definitions and expert-annotated examples in a **vector store** and retrieving the most relevant context per transcript segment to ground the LLM's annotations, with schema-constrained **JSON outputs**.
- Fusing interview annotations with **multimodal sensor data** (eye-tracking, 3D motion tracking) through temporal alignment, **segmentation and clustering** to characterise modes of exploratory activity.

**Casual Lecturer** *Sep 2023 – Jun 2026*
*Munster Technological University, Cork, Ireland*

- **MATH6031 Engineering Computing 1** (Mathematics Department) — computational mathematics and programming tools for engineering students.
- **Web Development** — front-end and back-end web technologies for undergraduate computing programmes.
- **Database Management Systems (DBMS)** — database design, normalisation, and MySQL; supervised student projects including the MTU Thoroughbreds Clinic system.
- **Data Structures & Algorithms** — algorithmic thinking, complexity analysis, and practical implementation.
- **Agile & Scrum Methodology** — taught Agile software development processes to Computer Science students; acted as **Scrum Master** for student project teams, facilitating sprints, backlog management using **Jira**, and iterative delivery.

**Data Analyst** *Mar 2025 – Oct 2026*
*Accenture, Dublin, Ireland*

- Data preprocessing, visualisation, and business insights using Python and SQL; cleaning and transforming raw data into structured formats; building interactive dashboards and visual reports to support strategic decision-making and present actionable insights to stakeholders.
- **Project 1 — AWS Data Pipeline:** Designed an end-to-end workflow for processing PDF and image data stored in AWS S3 using SNS and Lambda Functions, restoring extracted data back to S3 in cleaned format. Integrated **AWS Textract** and **Comprehend** for NLP-based text detection and classification. Implemented full automation using **AWS CDK in Python**, providing suitable triggers and event notifications to pass data across S3, SNS, and Lambda.
- **Project 2 — Azure-to-AWS Data Migration:** Designed the full workflow and architecture for enterprise data migration from Azure to AWS, acting as **Tech Lead and Data Engineer** for ETL processing. Used **AWS Glue** to build Staging, Data Warehouse, and DataMart tables in RDS MS-SQL; extracted data from JSON, Excel, and CSV sources, mapping into S3 and developing pipelines for filtering and loading into Amazon databases.

**Research Intern (NLP)** *Jan 2023 – present*

*GREYC Labs, Université Caen Normandie, France* — Advisor: Prof. Gaël Dias

- Primary focus: knowledge base-driven humour generation using Text Style Transfer, AMR graphs, Reinforcement Learning, and adversarial fine-tuning of **LLMs** (GPT-2, LLaMA, Phi, Mistral, Qwen).
- Introduced an adversarial learning method for LLMs to improve humour generation; contributed to the publication *"Generating Humorous Punchlines Through Adversarial Learning of Pretrained Language Models"*.
- Utilised the **CRIANN supercomputer cluster** (criann.fr) for large-scale parallel training and inference of LLMs across multiple GPU nodes for the project *"Corrective Feedback-Driven Humor Generation for Tiny Language Models"*.

**Research Intern (ML/DL)** *Mar 2022 – Oct 2022*
*ADAPT Centre, MTU, Ireland* — Advisor: Dr. Mohammed Hasanuzzaman

- Implemented "Hybrid Filter-Wrapper with Ensemble Classifier for Text Representations" on the diabetes behavioural dataset; evaluated on Reuters, Amazon Reviews, Book Reviews, 20 Newsgroups, and MT Samples datasets.
- **MSCA GenDAI Secondment** (Mar–Aug 2025): Competitively selected for a 6-month research secondment at **InConTec GmbH**, Pfungstadt, Germany, funded under **Marie Skłodowska-Curie Actions, Grant Agreement No. 101182801**, focusing on AI-powered genomic diagnostics.

## Publications

"Dynamic Model Switching to Mitigate Outdated Knowledge in Large Language Models." *LREC-2026*
*R. Pinninti (MTU), S. Kamila (Manipal Institute of Technology, Bengaluru), A. Mazumder (IBM, North Carolina, USA), M. Hasanuzzaman (ADAPT Centre, MTU).* ***Funded: MSCA Grant No. 101182801 & ADAPT Centre, SFI/ERDF, MTU.***

"Corrective Feedback-Driven Humor Generation for Tiny Language Models." ***Accepeted — AACL-IJCNLP 2026***
*R. Pinninti, L.C. Routhu, D. Bandyopadhyay, G. Dias, A. Lechervy, J. Pantin, M. Hasanuzzaman, A. Ekbal. Compute: CRIANN HPC Cluster, France.*

"Towards Building a Global Robust Model for Heart Disease Prediction." *ACM, 2023*
*B. Mali, S. Saha, D. Brahma, R. Pinninti, P.K. Singh.*

*Full publications:* Google Scholar

## Technical Skills

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| **Languages / Scripting** | Python, Java, C, SQL (MS-SQL, PostgreSQL, MySQL, BigQuery), Bash, JavaScript, Shell Scripting, API |
| **ML / Deep Learning** | Supervised & Unsupervised Learning, CNNs, RNNs, LSTMs, Transformers, Fine-tuning, Transfer Learning; PyTorch, TensorFlow, Keras, Scikit-learn |
| **LLMs & GenAI** | GPT-2, LLaMA, Phi, Mistral, Qwen, Gemma; LoRA/QLoRA, RLHF, Knowledge Distillation, RAG, HuggingFace, LangChain |
| **MLOps** | MLflow, W&B, Apache Airflow, Docker, CI/CD for ML, model versioning, pipeline orchestration, model serving |
| **HPC / Parallelisation** | Multi-GPU/multi-node (SLURM), **CRIANN Supercomputer** (criann.fr), vLLM, Ollama, DeepSpeed, distributed inference |
| **NLP** | HuggingFace, NLTK, spaCy, NumPy, SciPy, Pandas, Matplotlib |
| **Big Data** | Apache Hive, Impala, Hadoop, Apache Airflow |
| **Cloud — AWS** | S3, SNS, Lambda, Textract, Comprehend, Glue, CDK, Redshift, RDS, Cloud9, Boto3, Cognito, X-Ray |
| **Cloud — GCP / Azure** | BigQuery, Cloud Composer, Cloud Storage, Cloud Build, Cloud Data Fusion, MS Azure |
| **Agile / Project Mgmt** | Jira, Scrum, Sprint Planning, Backlog Management |
| **Tools / IDEs** | Jupyter, Colab, PyCharm, IntelliJ, Cloud Shell, iPython, Hue, Tableau, Git, LaTeX |

## Funding, Awards & Certifications

- **MSCA GenDAI Secondment** (Mar–Aug 2025): EU **Grant Agreement No. 101182801**, InConTec GmbH, Germany. cordis.europa.eu/project/id/101182801
- **ADAPT Centre Research Funding** (2022 – present): Supported by **Science Foundation Ireland (SFI)** and the **European Regional Development Fund (ERDF)**, Munster Technological University.
- Secured **88th percentile in JEE Mains 2019** (amongst 200,000 candidates).
- Received a prize from **BSNL** for best performance; designated as **Student Ambassador at Intel**.

**Certifications:**

- Machine Learning — Stanford University (Coursera)
- Neural Networks and Deep Learning — DeepLearning.AI
- Python for Everybody — University of Michigan
- AWS Fundamentals: Google Cloud Native
- Analyzing Big Data with SQL (Hive, Impala, SQL)
- Managing Big Data in Clusters and Cloud Storage (Hadoop, S3 & Big Data Management)
- Building Batch Data Pipelines on GCP

**References available on request.* *GitHub: github.com/Ramakrishna176420*